Algorithmic Pseudocode Sample 3

Source-backed Image2Struct algorithm sample

Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Algorithm 1 Initialization of the control and state variables

for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ *Retrieve the values of* $P_{\{u,t\}}$
  $P_{\{u,t\}} \leftarrow \texttt{Power.GetValue}(t)$
end for
for $t \in \{-1, ..., -T^{\text{traceback}}\}$ **do** ▷ *Initial conditions on the state variables*
  if $P_{\{u,t\}} > \texttt{MinimumPower.GetValue}(t)$ **then**
    $S^{\text{OFF}}_{\{u,t\}} \leftarrow 0$
    $S^{\text{STOP}}_{\{u,t\}} \leftarrow 0$
    $S^{\text{START}}_{\{u,t\}} \leftarrow 0$
    if $P_{\{u,t\}} < P_{\{u,t-1\}}$ **then** ▷ *Exact initialization required only if the FLAT state is defined.*
      $S^{\text{ON_UP}}_{\{t-1\}} \leftarrow 0$
      $S^{\text{ON_DOWN}}_{\{t-1\}} \leftarrow 1$
      $S^{\text{ON_FLAT}}_{\{t-1\}} \leftarrow 0$
    else if $P_{\{u,t\}} > P_{\{u,t-1\}}$ **then**
      $S^{\text{ON_UP}}_{\{t-1\}} \leftarrow 1$
      $S^{\text{ON_DOWN}}_{\{t-1\}} \leftarrow 0$
      $S^{\text{ON_FLAT}}_{\{t-1\}} \leftarrow 0$
    else if $P_{\{u,t\}} = P_{\{u,t-1\}}$ **then**
      $S^{\text{ON_UP}}_{\{t-1\}} \leftarrow 0$
      $S^{\text{ON_DOWN}}_{\{t-1\}} \leftarrow 0$
      $S^{\text{ON_FLAT}}_{\{t-1\}} \leftarrow 1$
    end if
  else if $P_{\{u,t\}} > 0$ **then** ▷ *Reconstruct the startups and shutdowns*
    if $P_{\{u,t\}} < P_{\{u,t-1\}}$ **then**
      $S^{\text{STOP}}_{\{t\}} \leftarrow 1$
      $S^{\text{START}}_{\{t\}} \leftarrow 0$
    else
      $S^{\text{STOP}}_{\{t\}} \leftarrow 0$
      $S^{\text{START}}_{\{t\}} \leftarrow 1$
    end if
  else ▷ *Final possibility: the unit is OFF.*
    $S^{\text{OFF}}_{\{u,t\}} \leftarrow 1$
    $S^{\text{STOP}}_{\{u,t\}} \leftarrow 0$
    $S^{\text{START}}_{\{u,t\}} \leftarrow 0$
    $S^{\text{ON_UP}}_{\{t-1\}} \leftarrow 0$
    $S^{\text{ON_DOWN}}_{\{t-1\}} \leftarrow 0$

$S_{\{t-1\}}^{\text{ON_FLAT}} \leftarrow 0$

end if

end for